Project Geographic Coordinates to Metric (km)

Description

Converts WGS84 lon/lat coordinates to planar metric coordinates in kilometres using a user-specified equal-area projection. The original lon/lat columns are retained alongside the projected 'coord_x_km' and 'coord_y_km' columns.

Usage

```
project_coords_to_metric(data_coords = NULL, target_crs = 3035L)
```

Arguments

`data_coords`	A data frame with 'dataset_name' as row names and columns 'coord_long' and 'coord_lat' (decimal degrees, WGS84), as returned by 'extract_dataset_coordinates()'.
`target_crs`	An integer EPSG code for the target projected coordinate reference system. Must be a metric (planar) equal-area CRS suitable for the geographic extent of the data. Defaults to '3035L' (ETRS89 Lambert Azimuthal Equal-Area, the standard for European analyses). For other regions, choose an appropriate equal-area projection, e.g.: * '6933L' — EASE-Grid 2.0 (global) * '5070L' — NAD83 Conus Albers (North America) * '102022L' — Africa Albers Equal Area Conic (Africa)

Details

Using metric coordinates instead of degrees is necessary for distance-based spatial modelling (e.g., Moran eigenvectors) because degree distances are not uniform across latitudes. The 'target_crs' default of EPSG:3035 is the standard equal-area projection for European spatial analyses; change it for non-European study regions.

The function requires the sf package to perform coordinate transformation. Dataset names (row names) are preserved unchanged in the output.

Value

A data frame with the same row names and columns as 'data_coords', plus two additional columns:

'coord_x_km'

Easting in kilometres (target CRS).

'coord_y_km'

Northing in kilometres (target CRS).

See Also

[extract_dataset_coordinates()], [prepare_spatial_predictors_for_fit()]